

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Collaborated with an *Agile* team of **10 engineers** to build a SaaS search platform serving **10K+ site visitors**
- Streamlined the search index by developing a de-duplication algorithm that flagged **30%** of documents as identical
- Optimized retrieval across **600K+ documents** in *Elasticsearch* to return search results in **under 1 second**

Product Development Intern | Jetson

January 2025 – February 2025

- Assessed **3 experimental features** via **7 iterations of UI/UX testing** on the Jetson App (**100K+ users**)
- Presented insightful recommendations to optimize user engagement in a **5-page report** delivered to Jetson's CEO
- Streamlined onboarding from **12 to 5 pages** in *Figma*, validating the redesign with **A/B testing** on 30 users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed **3 years** of TikTok data (**200+ posts**) with *Microsoft Excel*, *Power BI*, *Hootsuite* and *Python (pandas)*
- Found out that science “fun facts” posts made up just **7%** of all posted content yet earned **69.3%** of all views
- Synthesized the TikTok analysis findings into insightful, data-backed recommendations and a business case for a “fun facts” content strategy, pitched to the Executive Director, who adopted it, lifting engagement **24%** (8K+ views)

PROJECTS

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Automated scheduling with a **scheduling agent** (*React/Firebase*) that auto-books meetings on Google Calendar
- Designed a **5-step Python scheduling algorithm** that scans calendar availability, ranks the open time slots and emails a confirmation for each booking across large groups, validated across **14 unit and integration test cases**
- Signed **2 LOIs worth \$3.5K** and licensed Chronos to **8 startups**; **won 2025 Congressional App Challenge**
- Implemented a participant lobby where hosts authenticate invited participants and queue follow-up meetings

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led a team of **4** to **2nd at UIUC Founder's Hackathon** with a pricing algorithm for **Kamion (YC S22)**
- Cleaned and prepared data from **15,000** scraped truck-listing images via de-duplication and non-USD filtering
- Engineered a generative AI agent (*Claude Sonnet 4.6*) with a *JSON schema* to extract truck details from photos
- Tested and refined the extraction schema until **10 consecutive images** were correct on each of **3 features**
- Fit a *quantile regression* model on the extracted features to predict prices at **87% accuracy** across **30 samples**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Integrated **4 Random Forest** models into a *Python FastAPI* back end that monitors water, waste and energy data from **3 tools** to detect anomalies and deliver actionable steps via a *Gemma Voice LLM* on a *Next.js* dashboard
- Built a set of **3 edge AI tools** that track a school's energy, water and food consumption for anomaly detection
- Directed a team of **5** from initial ideation to the final presentation as a **USAII Global Hackathon Finalist**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Deployed a **RAG GenAI agent** (*Claude Sonnet 4.5*) that uses vector search to retrieve session-scoped documents
- Pulled papers from *OpenAlex* and filled missing metadata from *Semantic Scholar* in a two-API data pipeline
- Queried a *PostgreSQL* database hosted on *Supabase* with *SQL* to analyze papers by field, year and citation count
- Rendered each paper's outbound citations as an interactive *React Flow* graph spanning **26 research fields**

SKILLS

Programming Languages: Python, SQL, Java, JavaScript, C++, HTML/CSS

Frameworks/Platforms: Google Calendar API, Google Cloud, AWS, Supabase, PostgreSQL, Firebase, FastAPI, React

AI/ML: Claude API, Gemma 4, Hugging Face, Vercel AI SDK, Supermemory, scikit-learn, PyTorch, OpenCV

Data & Analytics: Microsoft Excel, Power BI, A/B Testing, pandas, NumPy, Hootsuite

Concepts/Practices: Process Optimization, Business Analysis, Business Case Development, Process Automation